CS661: Project Proposal Report

Visualizing the Impact of ENSO Phases on Indian Monsoon Rainfall

GROUP-22

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1 Introduction

The Indian Summer Monsoon is the backbone of India's agriculture economy, affecting the crop cycles, water security, etc. However, this is frequently disrupted by global climate anomalies, most notably the El Niño Southern Oscillation (ENSO) phases in the equatorial Pacific Ocean. A ENSO phase can lead to drought conditions in certain regions while leaving adjacent regions largely unaffected.

To truly understand the effect of such global climate anomalies. This project proposes an interactive, web based visual analytics system designed to map and understand relationships between oceanic shifts and regional land surface impacts. By utilizing long term historical baselines with high resolution satellite and meteorological datasets, our platform will provide a visualization driven framework to understand the effects of ENSO on Indian Monsoon. This system will leverage raw environmental data and show how such oceanic phenomena manifest as immediate changes on land.

2 Data Source

The project will integrate diverse meteorological, oceanographic, and satellite datasets to correlate global oceanic indices with regional land-surface impacts. The primary datasets include:

- **NOAA Extended Reconstructed Sea Surface Temperature (ERSST) v5:** A global monthly gridded SST dataset on a $2.0° \times 2.0°$ grid, dating back to 1854. This dataset is the official source for calculating the Oceanic Niño Index (ONI) to track ENSO phases.
- **NOAA Optimum Interpolation Sea Surface Temperature (OISST) v2.1:** A high-resolution ($0.25° \times 0.25°$) daily global SST dataset combining satellite and in-situ observations (available from 1981–present). It will be used for high-resolution comparison.
- **India Meteorological Department (IMD) High-Resolution Gridded Data:** Daily gridded rainfall ($0.25° \times 0.25°$) and temperature ($1.0° \times 1.0°$) datasets covering the Indian mainland, used to analyze daily monsoon progress, onset dates, and localized heatwave deviations.
- **Kaggle Historical Rainfall in India (1901–2015):** A historical baseline dataset compiled from district-wise and meteorological subdivision-wise rainfall records, used to calculate long-period averages (LPA) and establish historical context.
- **MODIS NDVI (Normalized Difference Vegetation Index):** Satellite-derived vegetation data at 1 km resolution, sourced from NASA's Terra/Aqua satellites. This dataset

will be used to analyze vegetation health and crop stress anomalies in regions experiencing monsoon delays.

3 Specific Tasks

To implement the visual analytics system, we will perform the following data processing and analysis tasks:

- **Data Aggregation & Gridding:** Crop and align spatial grids from different sources (IMD, NOAA, and MODIS) to common coordinate reference systems over the Indian subcontinent and the Niño 3.4 region.
- **Anomaly Computation:** Calculate Sea Surface Temperature Anomalies (SSTA) in the Pacific and Rainfall Deviations from the Long Period Average (LPA) over Indian meteorological subdivisions.
- **Temporal Index Alignment:** Construct running 3-month means of SST anomalies to match the ONI formulation and align them temporally with daily/monthly monsoon onset and rainfall indices.
- **Interactive Filtering & Spatial Querying:** Implement backend queries that allow users to filter data by year, month, state, or meteorological subdivision, and dynamically re-calculate correlations.
- **Agricultural Stress Mapping:** Compute historical and current vegetation anomaly indexes (NDVI deviations) to serve as a proxy for agricultural impact of rainfall.

4 Visualization Tasks

The interactive system will consist of six distinct visualization modules, each developed by our team to address a core visualization aspect of our data:

4.1 SST Dataset Resolution & Historical Coverage Comparison (OISST vs. ERSST)

It is imporatant to choose the right Sea Surface Temperature (SST) dataset for accurate ENSO tracking. ERSST provides a stable, long-term record (2.0° resolution, 1854–present) based on *in situ* observations, whereas OISST offers higher resolution (0.25° resolution, 1981–present) by combining satellite and *in situ* data, though this can introduce minor calibration biases. This visualization helps users to:

- Compare the grid density of ERSST and OISST side-by-side within the Niño 3.4 region (5°N–5°S, 120°W–170°W).
- Toggle between three major ENSO events : November 2015, November 2020, and November 2023, to see how ERSST's coarser grid filters out small-scale noise.
- Observe how OISST captures localized activity, such as eddies and upwellings, which confirms that ERSST is the better choice for calculating the Oceanic Niño Index (ONI).

Suggested Visualization: Synchronized side-by-side heatmaps featuring a diverging RdBu colorscale, a scenario dropdown menu, and hover tooltips for viewing exact SST values.

4.2 Oceanic Niño Index (ONI) Derivation and Interactive Analysis

Understanding the frequency, duration, and intensity of historical ENSO (El Niño-Southern Oscillation) events is crucial for climate analysis. This visualization will allow users to:

- Isolate specific temporal windows using a continuous timeline brush to analyze localized climate phases.
- Examine the exact mathematical derivation of the ONI by comparing raw Sea Surface Temperature (SST) against a 30-year base climatology on demand.
- Identify categorical phase distributions and uncover patterns in seasonal intensity across any selected timeframe.

Suggested Visualization: A vertically stacked Focus+Context diverging area chart acting as the primary anchor, triggering Details-on-Demand secondary charts (phase distribution donut, seasonal intensity heatmap, and derivation proof multi-line chart) housed in a sliding side panel.

4.3 Monthly Rainfall Distribution & Monsoon Progress Tracker

Understanding the intra-seasonal distribution and temporal progression of the monsoon is required to identify periods of peak rainfall dominance and tracking its cumulative advance over time. This visualization will allow users to:

- Track the bi-weekly progression of cumulative rainfall across the monsoon season to observe how the active trajectory stacks up against historical long-term averages.
- Analyze monthly rainfall distributions to isolate the onset timeline, periods of peak dominance, and prolonged break phases within the annual rainy season.
- Visualize the chronological advancement of the monsoon over time through an animated playback timeline to identify specific temporal shifts in regional distribution.

Suggested Visualization: A synchronized interactive dashboard featuring an interactive multi-line temporal progression chart, equipped with timeline animation playback controls, regional dropdown filters, and detailed hover tooltips showing monthly rainfall volume and cumulative anomalies.

4.4 Rainfall Deficits in El Niño vs. Non-El Niño Years

Understanding how ENSO phases influence India's monsoon variability is critical for assessing regional drought risk and rainfall anomalies. This visualization will allow users to:

- Compare percentage deviations from the Long Period Average (LPA) across Indian states and meteorological subdivisions for any two historical years.
- Switch between El Niño, La Niña, and Neutral years using a dual-year selector to directly observe spatial differences in monsoon performance.
- Quantify regional rainfall impacts through interactive hover tooltips displaying rainfall anomalies and the relative percentage difference between the selected years.

Suggested Visualization: Side-by-side synchronized plots of India with a diverging rainfall anomaly colorscale, dual-year dropdown selectors, and hover tooltips showing exact rainfall deviations and inter-year differences.

4.5 El Niño-Dependent Rainfall Correlation Analysis

Understanding the statistical relationship between ENSO intensity and regional rainfall variability is essential for quantifying the influence of Pacific Ocean warming on the Indian monsoon. This visualization will allow users to:

- Examine Pearson correlation coefficients between Oceanic Niño Index (ONI) values and regional rainfall anomalies across India.
- Compare rainfall sensitivity across different geographic regions using interactive region selectors and side-by-side correlation views.
- Investigate the strength and direction of ENSO-rainfall relationships through scatter plots with regression lines, enabling identification of regions most affected by El Niño and La Niña events.

Suggested Visualization: Interactive correlation heatmaps displaying Pearson correlation coefficients alongside synchronized scatter plots with regression fits, region selectors, and hover tooltips showing exact correlation values and rainfall statistics.

4.6 Vegetation & Agricultural Impact Mapping

In this part of the project, we look at the agricultural impact caused by El Niño monsoon delays, focusing on the crucial Kharif crop season (June–October). Through this module, users will be able to:

- Plot the national average NDVI directly against the Oceanic Niño Index (ONI) and break the data down by region (North, South, East, West, and Central) to pinpoint exactly where crops are most vulnerable.
- Toggle between major historical climate events like the 2009–2010 major drought or the strong 2015–2016 El Niño to see how vegetation stress changes geographically over time.
- Use time-window controls to filter specifically for the Kharif months. This helps isolate peak "browning" trends so we can clearly see the direct hit crop health takes when the monsoon is delayed mid-season.

Suggested Visualization: A dual-axis line chart for the national overview, linked up with regional correlation bar charts and multi-line anomaly graphs. We will also include interactive filters so users can easily select historical events and lock in the Kharif time window.

5 Overall Solution

Our proposed solution is an interactive, web-based visual analytics system. Instead of a static, single-page dashboard, the application is structured to tell a guided story from multiple perspectives. Users will start with a big-picture global view to explore Pacific ocean temperatures and ENSO trends, and then zoom in to see localized impacts like regional rainfall shortages and crop health.

The interface will feature a responsive layout with shared controls. For example, selecting a year or region in the main sidebar will instantly update all active charts. The visualizations are also fully linked: clicking on a specific district on the map will automatically refresh the corresponding rainfall timeline and vegetation graphs.

6 Tech Stack

To keep the application fast and highly interactive, we will use the following tools:

- **Backend:** Python (Django or FastAPI) to handle map-based queries, calculate real-time correlations, and send processed data to the user interface.
- **Data Processing:** Pandas, NumPy, and SciPy for statistical math, along with Rasterio and Geopandas to handle climate and satellite data grids.
- **Frontend:** HTML5, CSS, and JavaScript, using React.js for the page structure and D3.js or Plotly.js to build the interactive charts.
- **Database:** SQLite or PostgreSQL (with PostGIS extensions) to easily store and search geographic weather data

7 Team Members & Responsibilities

Each group member wil contribute to different aspects of the project.

- **Data Processing & Cleaning & Analytics:** Deepak Kumar (240329), Vishvas Patel(241179)
- **Backend Development:** Dev Prakash(240338), Shubham Kumar Patel(241010)
- **Visualisation Development:** Prakhar Gupta(240764), Sourabh Sankhala(241036).
- **Frontend Development:** Aayush Gajeshwar(240024)